Dollar Stablecoin and Arbitrage under Capital Controls

Evidence from the Kimchi Premium

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Master's Thesis

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Kimchi premium

- The kimchi premium refers to price discrepancies between Korean and offshore exchanges.
- Marginal investors outside the US and Europe are willing to pay more for bitcoin in response to positive news or sentiment.

$$Premium^{Kimchi} = \frac{P_{\mathrm{KR}}^{BTC/KRW}}{FX^{USD/KRW} \cdot P_{\mathrm{GLB}}^{BTC/USDT}} - 1$$

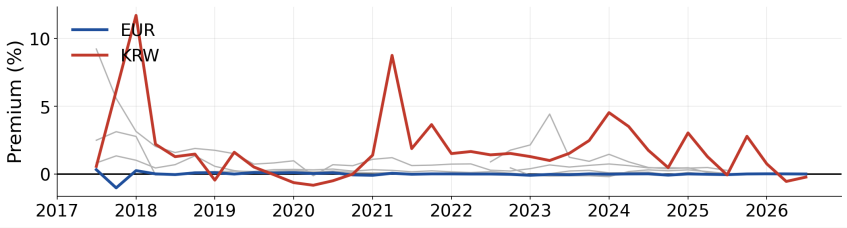
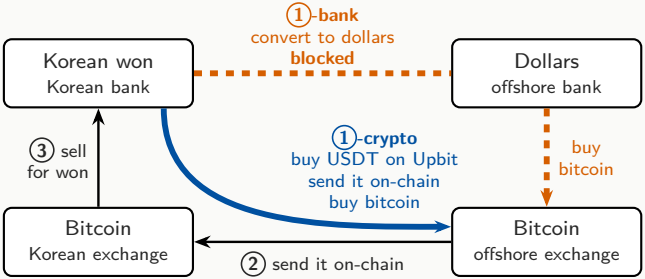


Figure 1: Bitcoin's premium in seven local currencies over Coinbase's USD price

Bitcoin's price on one local exchange per currency over Coinbase's USD price, quarterly mean of daily premia (Makarov & Schoar 2020). Grey: AUD, BRL, GBP, PHP, TRY, to 2025 Q3; KRW and EUR to 2026 Q3.

▸ Capital controls

Two routes for arbitrage: banking system or blockchain network



Makarov & Schoar (2020, *Journal of Financial Economics*)

*"[C]apital controls are most relevant for **the price of crypto to fiat currencies**. (**①-bank**)*
*In contrast, **transactions between two cryptocurrencies** (**①-crypto**) should not be affected [. . .]."*

Paying more for bitcoin, or for dollars?

- **USDT** is a cryptocurrency **pegged 1-to-1 to the US dollar** that **moves through blockchain**.
- Griffin & Shams (2020): stablecoins serve as the **medium for cross-exchange arbitrage**.

$$\underbrace{Premium^{Kimchi}}_{\text{kimchi premium}} \approx \underbrace{Premium^{Dollar}}_{\substack{\text{fiat-to-crypto spread} \\ \textbf{Dollar Demand}}} + \underbrace{Premium^{Crypto}}_{\substack{\text{crypto-to-crypto spread} \\ \textbf{Local Crypto Demand}}}$$

$$\frac{P_{\mathrm{KR}}^{BTC/KRW}}{\boldsymbol{FX}^{USD/KRW} \cdot P_{\mathrm{GLB}}^{BTC/USDT}} - 1 \approx \frac{P_{\mathrm{KR}}^{USDT/KRW}}{\boldsymbol{FX}^{USD/KRW}} - 1 + \frac{P_{\mathrm{KR}}^{BTC/KRW}}{P_{\mathrm{GLB}}^{BTC/USDT} \cdot \boldsymbol{P}_{\mathbf{KR}}^{\boldsymbol{USDT/KRW}}} - 1$$

Research question and Hypotheses

Research Question

Does USDT's value as an arbitrage vehicle push its price above one dollar?

H1 Vehicle demand. ($Premium^{Kimchi}$)
USDT trades above one dollar when more arbitrage runs through it.

H2 Capital controls. (across countries)
The dollar premium is larger where capital controls bind harder
and accounts for nearly all of the fiat premium (the kimchi premium, in Korea).

H3 Recycling capital. ($Premium^{Dollar}$)
Arbitrage runs on balances held in Korea: USDT deposits replenish what is sold above one dollar, while capital controls keep the won from being recycled below it.

Three strands of literature

- **Limits to arbitrage.** Identical assets can trade at different prices across markets because arbitrageurs face capital constraints and risk.
  Shleifer & Vishny (1997, *Journal of Finance*); Froot & Dabora (1999, *Journal of Financial Economics*); Gromb & Vayanos (2002, *Journal of Financial Economics*)

- **The kimchi premium and capital controls.** Crypto premia are largest across borders with capital controls, which bind the fiat-to-crypto price; crypto also carries capital flight and remittances across them.
  Makarov & Schoar (2020, *Journal of Financial Economics*); Choi, Lehar & Stauffer (2022, working paper, R&R *Journal of Financial and Quantitative Analysis*); Graf von Luckner, Reinhart & Rogoff (2023, *Journal of Monetary Economics*)

- **Stablecoins in crypto markets.** Stablecoins serve as a common numeraire and an arbitrage vehicle across crypto exchanges, and arbitrage holds their dollar peg.
  Griffin & Shams (2020, *Journal of Finance*); Lyons & Viswanath-Natraj (2023, *Journal of International Money and Finance*); Kang, Kang, Kim & Sul (2025, *Economics Letters*)

Contribution

1. Few papers explain why USDT trades above one dollar; we show that **demand for USDT as an arbitrage vehicle** pushes its price above its one-dollar fundamental.

2. This dollar premium is larger where **capital controls** and other frictions on arbitrage bind harder, and it accounts for nearly all of the fiat premium (the kimchi premium, in Korea).

3. Tether's issuance pulls the price back to one dollar, so the premium prices **USDT's service as a vehicle** until supply catches up.

Empirical Design: Recycling Capital (H3)

x: cryptoasset, t: 10 minutes

Order book $NetSide_{x,t} = \sum BuyerInit_{x,t} - \sum SellerInit_{x,t}, \quad NSell_{x,t} = -NetSide_{x,t}$

Wallets $NIn_{x,t} = In_{x,t} - Out_{x,t}$ at the venue's own wallets

$$\frac{1}{3}\sum_{k=0}^{2} \Delta Premium^{Dollar}_{t+k} = \alpha + \boldsymbol{b}\, NIn_{USDT,t-1} + \gamma' W^{D}_{t-1} + \delta\, 1[FeeDC]_t + \varepsilon_t$$

Prediction

$\boldsymbol{b} < 0$: USDT brought in to sell lowers $Premium^{Dollar}$.

Specification

The kimchi premium's swings already sit in USDT's and BTC's own order flow

$NetSide_{x,t} = \sum BuyerInit_{x,t} - \sum SellerInit_{x,t}$, daily, at Upbit, in BTC.

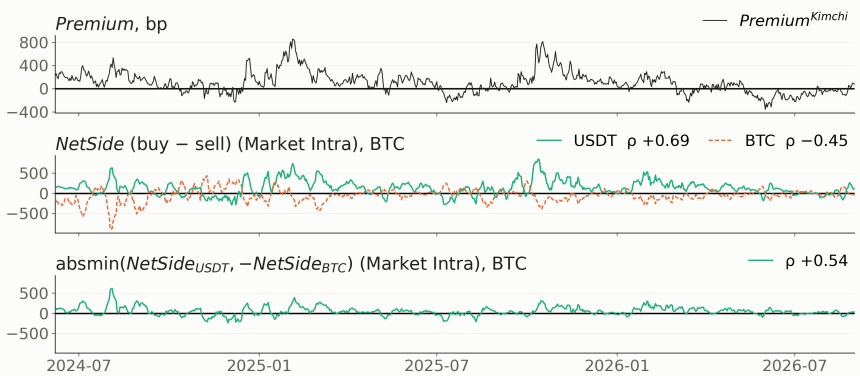


Figure 2: USDT's and BTC's flows at Upbit, and the kimchi premium

Upbit, daily, 7-day centered means, 10 Jun 2024 – 31 Aug 2026; each minute's won over that minute's BTC/KRW.

After the USDT/KRW listing, 10 Jun. 2024 ∼ 31 Aug. 2026 (%)

	BTC	ETH	SOL	XRP	DOGE	TRX	LINK	AVAX	SAND	SHIB	ADA	DOT	ALGO	BCH	ETC	STX	POL	USDT
BTC	100																	
ETH	23	100																
SOL	38	44	100															
XRP	2	41	25	100														
DOGE	-2	26	23	39	100													
TRX	-13	3	0	30	1	100												
LINK	8	32	20	44	31	16	100											
AVAX	8	24	32	41	38	13	41	100										
SAND	-3	12	9	46	37	38	33	34	100									
SHIB	9	20	28	25	56	-3	18	36	29	100								
ADA	11	30	25	46	26	9	29	23	24	14	100							
DOT	-8	12	7	32	30	15	20	22	51	31	21	100						
ALGO	-7	13	15	50	31	49	33	41	55	29	32	30	100					
BCH	4	13	12	6	24	17	8	22	13	36	6	20	14	100				
ETC	5	24	13	30	38	7	33	33	34	37	14	30	25	48	100			
STX	32	22	39	20	39	4	25	36	31	60	9	21	19	33	40	100		
POL	-6	21	14	53	47	37	43	48	69	53	31	54	56	39	60	63	100	
USDT	-51	-59	-52	-36	-17	2	-25	-23	0	-8	-23	-3	-2	-6	-11	-19	0	100

Before the listing

USDT is also the one crypto whose own flow tracks the kimchi premium, but only since it started trading

Year-long windows, 1 Jun. 2022 ∼ 30 Sep. 2026 (%)

	USDT	SOL	BTC	ETH	XRP	STX	LINK	AVAX	ADA	DOGE	SHIB	TRX	SAND	ETC	ALGO	POL	DOT	BCH
2022-06 – 2023-05		-34	-28	-35	0	-3	-24	-4	-13	-17	-5	16	-28	-13	-26		-10	-4
2023-06 – 2024-05		-26	-25	-23	8	3	-2	-35	-19	-22	-45	-21	-6	-15	-13		-19	13
2024-06 – 2025-05	61	-35	-36	-24	-12	-4	-9	1	-8	3	11	11	17	8	11	24	12	11
2025-06 – 2026-05	51	-27	-33	-34	-22	-7	-10	-27	-12	-19	-2	0	-14	4	12	-2	-7	1
2026-06 – 2026-09	25	-1	-8	18	-7	-21	2	5	14	13	-6	-10	11	-13	19	-7	-9	2

The dollar premium is larger where finance is less free

			$Premium_{BTC}$		$Premium_{USDT}$	
	Bins	Financial Freedom	Mean	SD	Mean	SD
AUD (BTC Markets)	47,092	80	1.5	23.8	-0.6	26.0
GBP (Coinbase)	116,026	80	-1.1	15.9	-3.3	16.1
EUR (Kraken)	116,689	70	0.8	12.4	-1.4	12.2
CAD (Kraken)	62,040	70	-3.1	30.2	-5.9	27.8
KRW (Upbit)	116,726	60	105.6	181.2	103.2	180.6
TWD (MAX)	115,235	60	45.8	42.6	44.0	41.3
PHP (Coins.ph)	116,548	60	20.0	30.0	20.1	27.6
TRY (BtcTurk)	116,111	60	11.4	42.7	10.2	40.9
MXN (Bitso)	113,700	60	6.6	35.3	5.0	32.4
BRL (Foxbit)	100,971	40	33.9	43.1	33.1	42.0

▸ Korea's capital controls

Dollar premia move together less where markets are less open

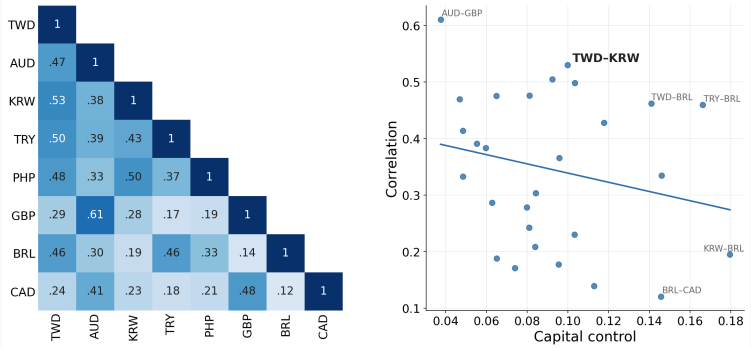


Figure 3: Pairwise correlations of the dollar premium across fiat currencies

$Premium^{Dollar}$: daily means of Table 3's bins but EUR's and MXN's, 30 days a pair at least. Capital control: the product of each country's 1 − Open Markets/100 (Heritage 2026's Trade, Investment and Financial Freedom, averaged); matrix by mean correlation; tagged: the one in bold and the 5 with the largest Cook's distance. OLS -0.82 ($t = -1.17$, $R^2 = 0.05$, 28 pairs); Makarov & Schoar (2020) Fig. 9, bitcoin: $+0.29$ ($t = 3$).

▸ NSell's effect by country

Arbitrage runs on balances held on both sides

Makarov & Schoar (2020, *Journal of Financial Economics*), Section 8.1

"The second arbitrage strategy is to hold a positive balance of bitcoins on both exchanges [. . .]. To replenish it, the arbitrageur needs to transfer bitcoins from the exchange with high bitcoin balance to the one with low balance and vice versa for capital. [. . .] Arbitrage becomes more efficient the quicker the arbitrageur can recycle capital."

- **USDT above one dollar:** arbitrageurs sell USDT held at Upbit; deposits replenish it.
- **USDT below one dollar:** arbitrage spends won held at Upbit, which capital controls keep from being recycled across the border.

USDT deposits lower the dollar premium above one dollar

$$\frac{1}{3}\sum_{k=0}^{2}\Delta Premium_{t+k}^{Dollar} = \alpha + \sum_{x} \boldsymbol{b_x}\, NIn_{x,t-1} + \gamma' W_{t-1}^{D} + \delta\, 1[FeeDC]_t + \varepsilon_t$$

	Panel A: All			Panel B: $Premium_{t-1} > +1\%$		
	(1) Full	(2) Lag Neg	(3) Lag Pos	(4) Full	(5) Lag Neg	(6) Lag Pos
$NIn_{USDT,t-1}$	-0.159***	-0.148***	-0.167***	-0.247***	-0.227***	-0.258***
	(-4.35)	(-3.89)	(-2.93)	(-4.32)	(-4.45)	(-2.82)
$NIn_{BTC,t-1}$	-0.024	-0.040	+0.000	-0.048	-0.042	-0.044
	(-0.82)	(-0.86)	(+0.01)	(-0.95)	(-0.54)	(-0.77)
$NIn_{ETH,t-1}$	-0.086**	-0.109***	-0.061	-0.131**	-0.182***	-0.080
	(-2.52)	(-2.60)	(-1.39)	(-2.45)	(-3.28)	(-1.08)
$NIn_{XRP,t-1}$	+0.092	+0.145	+0.088	+0.108	+0.139	+0.127
	(+1.00)	(+1.16)	(+0.87)	(+0.72)	(+0.71)	(+0.70)
$NIn_{DOGE,t-1}$	-0.124*	-0.162	-0.087	-0.214**	-0.283*	-0.149
	(-1.70)	(-1.50)	(-1.45)	(-2.01)	(-1.85)	(-1.62)
σ_{t-1}	-0.019	-0.088***	+0.077***	-0.020	-0.086***	+0.073**
	(-0.90)	(-3.63)	(+2.87)	(-0.83)	(-2.96)	(+2.27)
$\Delta Premium_{t-1}$	-0.059***	-0.052***	-0.040***	-0.042***	-0.036**	-0.032
	(-7.28)	(-4.73)	(-3.01)	(-4.20)	(-2.22)	(-1.60)
$\Delta Premium_{t-1}\sigma_{t-1}$	-0.001	-0.001	-0.002***	-0.001*	-0.001	-0.002***
	(-1.32)	(-1.34)	(-3.31)	(-1.69)	(-1.62)	(-3.14)
$FeeDC_{j,t}$	+0.038	-0.186***	+0.266***	+0.059	-0.266***	+0.365***
	(+0.90)	(-3.46)	(+4.85)	(+0.82)	(-2.96)	(+4.01)
Constant	+0.139	+0.912***	-0.822***	+0.028	+0.828***	-0.915***
	(+0.86)	(+4.63)	(-3.50)	(+0.14)	(+3.32)	(-2.99)
Observations	116,626	50,802	50,000	54,720	23,329	23,811
Adjusted R^2	+0.025	+0.017	+0.024	+0.026	+0.022	+0.033

USDT deposits lower the dollar premium at Upbit, MAX, BtcTurk and Coinbase

$$\frac{1}{3}\sum_{k=0}^{2}\Delta Premium^{Dollar}_{e,t+k} = \alpha_e + \boldsymbol{b}\, NIn_{USDT,e,t-1} + \gamma' W^{D}_{e,t-1} + \delta\, 1[FeeDC]_t + \varepsilon_t$$

	10 min				
	(1) Issuance	(2) No Issuance	(3) Lag Neg	(4) Lag Pos	(5) Lag Neg, Issuance
Panel A: Tether Treasury issues into					
$NIn_{USDT,t-1}$	-0.124**	-0.035***	-0.053***	-0.041**	-0.055
	(-2.00)	(-5.38)	(-5.01)	(-2.47)	(-1.32)
σ_{t-1}	-0.033	-0.000	-0.054***	+0.051***	-0.096**
	(-1.13)	(-0.02)	(-3.32)	(+3.31)	(-2.43)
$\Delta Premium_{t-1}$	-0.026	-0.070***	-0.062***	-0.059***	-0.042*
	(-1.23)	(-7.89)	(-4.94)	(-2.99)	(-1.86)
$\Delta Premium_{t-1}\sigma_{t-1}$	-0.002*	-0.000	-0.001	-0.002	-0.001
	(-1.69)	(-0.16)	(-0.74)	(-1.43)	(-0.87)
$FeeDC_{j,t}$	+0.165	+0.001	-0.241***	+0.298***	-0.149
	(+1.39)	(+0.03)	(-4.45)	(+5.01)	(-1.09)
Exchange FE	Yes	Yes	Yes	Yes	Yes
Exchanges	4	4	4	4	4
Observations	44,976	426,649	200,490	203,899	20,079
Within R^2	+0.027	+0.020	+0.014	+0.021	+0.024
Panel B: no issuance					
$NIn_{USDT,t-1}$		-0.013**	-0.014*	-0.012	
		(-2.42)	(-1.88)	(-1.19)	
σ_{t-1}		+0.006	-0.020***	-0.002	
		(+0.65)	(-2.77)	(-0.38)	
$\Delta Premium_{t-1}$		-0.121***	-0.115***	-0.121***	
		(-17.58)	(-6.10)	(-10.12)	
$\Delta Premium_{t-1}\sigma_{t-1}$		-0.000	-0.001	+0.000*	
		(-0.31)	(-1.36)	(+1.90)	
Exchange FE		Yes	Yes	Yes	
Exchanges		5	5	5	
Observations		591,655	230,301	230,284	
Within R^2		+0.075	+0.152	+0.050	

Conclusion: Where Each Hypothesis Stands

	Premium	Evidence	Finding	Status
H1	$Premium^{Kimchi}$	Fig. 2	USDT bought, BTC sold as $Premium^{Kimchi}$ rises	Stylized fact
	$Premium^{Kimchi}$	Table 1	every crypto's *NetSide* comoves but USDT's	Stylized fact
	$Premium^{Kimchi}$	Table 2	USDT's *NetSide* alone tracks $Premium^{Kimchi}$	Stylized fact
H2	$Premium^{Dollar}$, $Premium^{Kimchi}$	Table 3	$D \approx K$, larger where finance is less free	Supported
H3	$Premium^{Dollar}$	Table 4	deposits replenish above \$1, no recycling below	Supported
	$Premium^{Dollar}$	Table 5	NIn_{USDT} lowers $Premium^{Dollar}$ at 4 exchanges	Supported

References

Choi, Kyoung Jin, Alfred Lehar, and Ryan Stauffer. 2022. "Bitcoin Microstructure and the Kimchi Premium." Working paper, University of Calgary.

Duffie, Darrell. 2010. "Presidential Address: Asset Price Dynamics with Slow-Moving Capital." *Journal of Finance* 65 (4): 1237–67.

Evans, Martin D. D., and Richard K. Lyons. 2002. "Order Flow and Exchange Rate Dynamics." *Journal of Political Economy* 110 (1): 170–80.

Froot, Kenneth A., and Emil M. Dabora. 1999. "How Are Stock Prices Affected by the Location of Trade?" *Journal of Financial Economics* 53 (2): 189–216.

Graf von Luckner, Clemens, Carmen M. Reinhart, and Kenneth Rogoff. 2023. "Decrypting New Age International Capital Flows." *Journal of Monetary Economics* 138: 104–22.

Griffin, John M., and Amin Shams. 2020. "Is Bitcoin Really Untethered?" *Journal of Finance* 75 (4): 1913–64.

Gromb, Denis, and Dimitri Vayanos. 2002. "Equilibrium and Welfare in Markets with Financially Constrained Arbitrageurs." *Journal of Financial Economics* 66 (2–3): 361–407.

Kang, Chang-Mo, Hyoung-Goo Kang, Donghyun Kim, and Hong Kee Sul. 2025. "Stablecoin and Cross-Border Crypto Market Integration." *Economics Letters* 257: 112704.

Kyle, Albert S. 1985. "Continuous Auctions and Insider Trading." *Econometrica* 53 (6): 1315–35.

Lyons, Richard K., and Ganesh Viswanath-Natraj. 2023. "What Keeps Stablecoins Stable?" *Journal of International Money and Finance* 131: 102777.

Makarov, Igor, and Antoinette Schoar. 2020. "Trading and Arbitrage in Cryptocurrency Markets." *Journal of Financial Economics* 135 (2): 293–319.

Shleifer, Andrei, and Robert W. Vishny. 1997. "The Limits of Arbitrage." *Journal of Finance* 52 (1): 35–55.

Appendix

Data Sources

USDT/fiat books in 20 currencies at **Upbit**, **Coinbase** and venues abroad; **BTC/fiat books** in 13 currencies, against Coinbase's BTC/USD.

Data	Frequency	Provider	Coverage	Period
USDT/fiat prices	Minute	Upbit, MAX, BtcTurk, Kraken, Coinbase, BTC Markets, Binance, Bitso, Foxbit, Coins.ph, HashKey, VALR	31 books, 20 currencies	2024-06 – 2026-09
BTC/fiat prices	Minute, daily	Upbit, MAX, BtcTurk (minute); Kraken, Coinbase, BTC Markets, Foxbit, Coins.ph, Bitso, VALR, Coinmate, bitbank	13 currencies; BTC/USD at Coinbase	2024-06 – 2026-09
FX rate	Minute	Barchart	USD against each currency	2024-06 – 2026-09
Trades	Tick, minute	Upbit, MAX, Kraken, Coinbase, Bitso, BTC Markets, VALR (ticks); Binance, Foxbit, Coins.ph, HashKey (taker volume)	USDT's *NetSide*	2024-06 – 2026-09
Transfers	Transaction	Arkham Intelligence, XRP Ledger, Blockchain.com, TronGrid, Tenderly, Alchemy (Flipside)	Upbit's USDT, BTC, ETH, XRP, DOGE; USDT at 7 venues; Tether Treasury to venues	2024-06 – 2026-09
Index	Annual	Heritage Foundation	Financial Freedom, 2026 edition	2025

Variable	Definition
$Premium_t^{Kimchi}$	$P_t^{BTC/KRW} / \left(FX_t^{USD/KRW} \cdot P_{GLB,t}^{BTC/USDT}\right) - 1$
$Premium_t^{Dollar}$	$P_t^{USDT/KRW} / FX_t^{USD/KRW} - 1$
$Premium_t^{Crypto}$	$P_t^{BTC/KRW} / \left(P_{GLB,t}^{BTC/USDT} \cdot P_t^{USDT/KRW}\right) - 1$
$NetSide_{x,t}$, $NSell_{x,t}$	$\sum BuyerInit_{x,t} - \sum SellerInit_{x,t}$ on the venue's fiat book (Upbit's KRW, Coinbase's USD), $NSell_{x,t} = -NetSide_{x,t}$
$\Delta Reserve_{x,t}$, $NIn_{x,t}$	$Out_{x,t} - In_{x,t}$ at the venue's own wallets (Upbit's; MAX's and BtcTurk's USDT), $NIn_{x,t} = -\Delta Reserve_{x,t}$
σ_t	standard deviation of $\Delta Premium^{Dollar}$ over the 12 hours to t
$FeeDC_t$	1 during Upbit's USDT/KRW fee discount event, 0 otherwise
Issuance	the 72 hours after a Tether Treasury transfer of 1M USDT or more into the venue

Recycling Capital (H3): Specification

- *NIn* in BTC, a minute's net under \$1,000 counted as zero, standardized within 14-month blocks.
- $W^{D}_{t-1} = \left(\sigma_{t-1},\ \Delta Premium^{Dollar}_{t-1},\ \Delta Premium^{Dollar}_{t-1}\,\sigma_{t-1}\right)'$, σ_t: s.d. of $\Delta Premium^{Dollar}$ over the 12 hours to t; $1[FeeDC]_t$: Upbit's USDT/KRW fee discount.
- Columns as Griffin & Shams (2020) Table II: every bin, then the bins after a fall or a rise in $Premium^{Dollar}$.
- Newey-West errors, 3 lags.

◂ Empirical design

At MAX, USDT deposits do not lower the dollar premium

	MAX USDT/TWD		
	(1) Full	(2) Lag Neg	(3) Lag Pos
$NIn_{USDT,t-1}$	+0.007	+0.010	+0.007
	(+0.82)	(+0.68)	(+0.54)
σ_{t-1}	-0.012	-0.014	+0.016
	(-0.81)	(-0.74)	(+0.72)
$\Delta Premium_{t-1}$	-0.062***	-0.093***	-0.058*
	(-5.86)	(-6.63)	(-1.79)
$\Delta Premium_{t-1}\sigma_{t-1}$	+0.002*	+0.004***	-0.000
	(+1.86)	(+2.75)	(-0.12)
Constant	+0.046	-0.064	+0.037
	(+0.89)	(-0.96)	(+0.44)
Observations	118,036	51,325	52,487
Adjusted R^2	+0.007	+0.012	+0.008

Before the USDT/KRW listing, 1 May 2022 ∼ 31 May 2024 (%)

	BTC	ETH	SOL	XRP	DOGE	TRX	LINK	AVAX	SAND	SHIB	ADA	DOT	ALGO	BCH	ETC	STX
BTC	100															
ETH	43	100														
SOL	33	27	100													
XRP	11	10	13	100												
DOGE	21	24	22	21	100											
TRX	11	12	13	9	2	100										
LINK	19	26	30	12	22	4	100									
AVAX	9	7	19	5	5	-4	20	100								
SAND	29	35	32	18	28	21	26	18	100							
SHIB	9	5	9	6	31	14	6	9	5	100						
ADA	36	46	15	19	16	26	31	19	39	12	100					
DOT	14	33	23	8	11	10	32	32	30	14	32	100				
ALGO	17	28	40	27	42	10	23	5	30	5	29	21	100			
BCH	-1	-1	-3	3	4	-5	2	-2	-3	8	5	3	-4	100		
ETC	20	43	9	3	7	8	9	13	26	21	35	27	15	12	100	
STX	9	-2	-1	3	5	6	5	-1	16	-1	1	5	-3	14	2	100

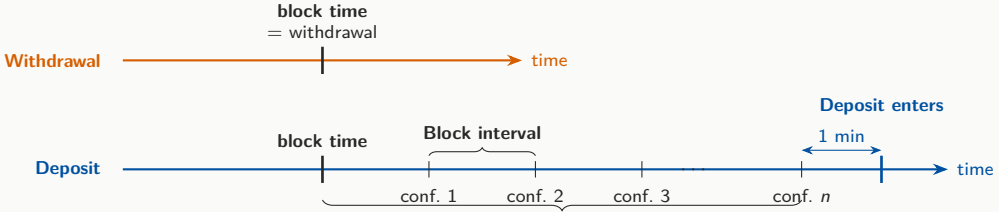
◂ Table 1

Tether Treasury's direct transfers with Panel B's exchanges

	From Treasury				To Treasury	
	Transfers	$\geq$ 1M	USDT (M)	Largest (M)	Transfers	USDT (M)
BTC Markets	0	0	0.00	–	0	0.00
VALR	0	0	0.00	–	0	0.00
Bitso	0	0	0.00	–	0	0.00
Coins.ph	0	0	0.00	–	0	0.00
HashKey	0	0	0.00	–	0	0.00
No label names the address	1,700	655	9,247.84	750.00	298	3,211.16

When Each Transfer Enters the Flows

	Upbit confirmations	Block interval	Deposit enters
BTC	1	10 min	next block's time + 1 min
ETH	36	12 s	36th block on top's time + 1 min
USDT (Tron)	1	3 s	block time + 1.05 min
USDT (Ethereum)	36	12 s	36th block on top's time + 1 min
XRP	1	4 s	block time + 1.07 min
DOGE	6	1 min	6th block on top's time + 1 min
SOL	1	0.4 s	block time + 1.01 min



Upbit confirmations n: blocks mined on top (BTC, DOGE, ETH: the confirming block's own time)

As the dollar premium falls, USDT comes in and cryptos go out

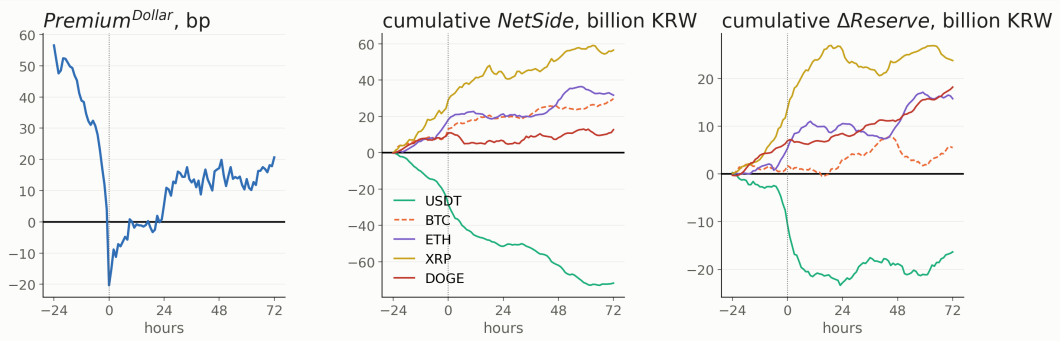


Figure A1: Upbit's flows around a fall of the dollar premium

Upbit, hourly, 8 Jun 2024 – 31 Aug 2026: the 36 hours $Premium^{Dollar}$ falls to −10 bp or below after 24 hours above it, at hour 0. Each flow less its own mean at that UTC hour, averaged over the events and summed from hour −24; $\Delta Reserve$ with the other Korean venues.

NSell's effect on the dollar premium, by USDT restriction

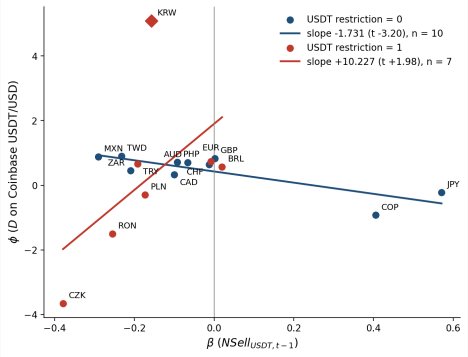


Figure A2: NSell's effect on $Premium^{Dollar}$ against $Premium^{Dollar}$'s pass-through from Coinbase's USDT/USD

A country's USDT/fiat book, 10-minute, Jun 2024 – Aug 2026. β: its $NSell_{USDT,t-1}$ coefficient; ϕ: the slope of its D on Coinbase's USDT/USD D, each winsorized at 1% and 99%. Restricted: Turkey, Brazil, Korea, EU members. Argentina and Coinbase left out.

◂ Figure 3 ▸ Table of the fits

Panel A: a USDT restriction

	BRL (Bitso)	KRW (Upbit)	TRY (Binance)	RON (Binance)	CZK (Binance)	EUR (Kraken)	PLN (Binance)
$NSell_{USDT,t-1}$	+0.020	-0.157***	-0.192***	-0.255***	-0.379***	-0.008*	-0.173**
	(+0.79)	(-3.31)	(-13.26)	(-3.98)	(-5.62)	(-1.85)	(-2.32)
$Nlin_{USDT,t-1}$	-0.021	-0.111***					
	(-1.62)	(-3.38)					
σ_{t-1}	-0.031	-0.029	+0.014	+0.021**	+0.024	+0.031	+0.032
	(-1.32)	(-1.23)	(+0.97)	(+2.00)	(+1.23)	(+1.14)	(+0.71)
$\Delta Premium_{t-1}$	-0.082***	-0.063***	-0.060***	-0.116***	-0.124***	-0.083***	-0.079***
	(-5.74)	(-7.97)	(-4.80)	(-12.69)	(-13.26)	(-8.55)	(-3.46)
$\Delta Premium_{t-1}\sigma_{t-1}$	-0.000	-0.001	-0.001	-0.000	-0.000	+0.002	+0.000
	(-0.96)	(-1.31)	(-0.86)	(-0.84)	(-0.08)	(+1.06)	(+0.17)
$FeeDC_{i,t}$		+0.039					
		(+0.91)					
Constant	+0.218	+0.226	-0.051	-0.198**	-0.259	-0.046	-0.199
	(+1.40)	(+1.21)	(-1.05)	(-2.27)	(-1.35)	(-1.20)	(-0.77)
Observations	114,728	116,632	118,331	42,145	35,706	118,026	108,920
Adjusted R^2	+0.084	+0.025	+0.023	+0.069	+0.084	+0.020	+0.026

Panel B: no USDT restriction

	ZAR (VALR)	TWD (MAX)	MXN (Binance)	COP (Bitso)	PHP (Coins.ph)	JPY (Kraken)	ARS (Binance)	CAD (Kraken)
$NSell_{USDT,t-1}$	-0.209***	-0.232***	-0.291***	+0.406***	-0.092***	+0.571	-0.111**	-0.100***
	(-19.58)	(-11.21)	(-12.58)	(+2.93)	(-6.13)	(+1.45)	(-2.11)	(-2.59)
$Nlin_{USDT,t-1}$	-0.043***	+0.010		+0.015	-0.003			
	(-4.61)	(+1.12)		(+0.27)	(-0.46)			
σ_{t-1}	+0.001	-0.018	+0.007	+0.002**	+0.015	+0.010	-0.001	+0.026
	(+0.12)	(-1.17)	(+0.77)	(+2.22)	(+0.90)	(+0.64)	(-0.08)	(+0.66)
$\Delta Premium_{t-1}$	-0.111***	-0.080***	-0.111***	-0.059*	-0.103***	-0.064***	-0.043***	-0.024
	(-12.82)	(-7.56)	(-27.99)	(-1.78)	(-4.36)	(-11.39)	(-5.03)	(-1.02)
$\Delta Premium_{t-1}\sigma_{t-1}$	+0.001	+0.002**	-0.000	+0.000*	-0.000	+0.000	+0.001*	+0.001***
	(+0.85)	(+2.30)	(-1.18)	(+1.79)	(-0.48)	(+0.73)	(+1.71)	(+2.69)
Constant	-0.014	+0.066	-0.064	-0.004	-0.068	-0.061	-0.020	-0.092
	(-0.45)	(+1.28)	(-9.86)	(-0.08)	(-0.93)	(-0.50)	(-0.25)	(-0.85)
Observations	112,956	118,036	117,712	67,885	118,287	14,365	118,331	63,765
Adjusted R^2	+0.052	+0.015	+0.064	+0.070	+0.085	+0.076	+0.003	+0.017

Panel C: no USDT restriction

	GBP (Kraken)	AUD (Kraken)	CHF (Kraken)	USD (Coinbase)
$NSell_{USDT,t-1}$	+0.002	-0.066***	-0.012	-0.005***
	(+0.61)	(-9.73)	(-0.61)	(-3.12)
$Nlin_{USDT,t-1}$				-0.006***
				(-3.75)
σ_{t-1}	+0.040**	+0.030**	+0.030*	-0.018
	(+1.96)	(+2.06)	(+1.74)	(-1.20)
$\Delta Premium_{t-1}$	-0.123***	-0.120***	-0.147***	-0.110***
	(-13.86)	(-11.78)	(-8.85)	(-13.17)
$\Delta Premium_{t-1}\sigma_{t-1}$	+0.006**	+0.001	+0.010***	+0.002
	(+2.24)	(+0.40)	(+3.62)	(+0.20)
Constant	-0.073**	-0.080**	-0.090**	+0.009
	(-2.11)	(-2.16)	(-2.00)	(+1.26)
Observations	117,772	113,901	107,998	118,249
Adjusted R^2	+0.042	+0.065	+0.040	+0.057

◂ Figure A2

Crypto Share of Upbit's Trading Value

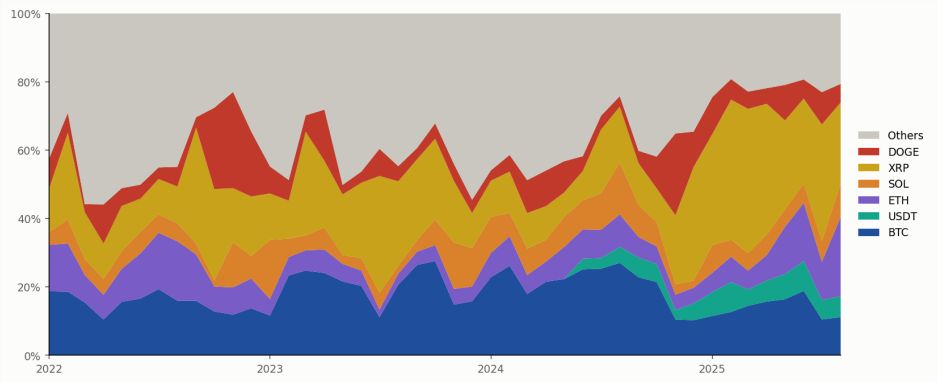


Figure A3: Crypto share of Upbit's monthly KRW trading value, Jan. 2022 – Aug. 2025

Months in UTC, from the one-minute candles, price × volume. Others: the other KRW books held; the figure stops in Aug. 2025, the last month before Upbit's books held fall from 36 to 18.

Korea won's capital controls

Rule	In force	Source
Remittance cap	A resident may send abroad without documents at most **USD 50,000** a year (1999 to 3 July 2023), **USD 100,000** a year since 4 July 2023, and **USD 5,000** per transfer (from Jan 2026)	Ministry of Economy and Finance (10 Feb 2023; 8 Dec 2025)
Exchange access	Won deposits require a real-name bank account; **non-residents cannot trade** on Korean exchanges (since 30 Jan 2018)	Financial Services Commission (23 Jan 2018)
Enforcement	**USD 12.26 billion** of outward remittances by 84 firms, collected from crypto exchanges and sent abroad as import payments, 2021–22	Financial Supervisory Service (4 Apr 2023)

◂ Kimchi premium

Country	Annual limit per individual	Changes, Jun 2024 – Sep 2026	Source
Panel A: a limit or a change			
Argentina	USD 200 a month (about USD 2,400 a year)	14 Apr 2025: no limit on transfers from a local bank account; Sep 2025: 90-day restriction extended to all individuals	BCRA A 8226, A 8336
South Africa	R1 million (about USD 55,000); above it, Reserve Bank approval	8 Apr 2026: raised to R2 million (about USD 110,000)	SARB Circular 6/2026
Taiwan	USD 5 million; above it, central bank approval	1 Nov 2024: raised to USD 10 million	CBC press release, 30 Oct 2024
South Korea	USD 100,000 without documents	29 Dec 2025: one limit across all institutions	MOEF Regulation Art. 4-3
Philippines	None on current transfers; USD 60 million for outward investment	None found	BSP FX Manual
Brazil	None	2 Feb 2026: payments abroad in crypto through non-bank providers capped at USD 100,000	Law 14,286, BCB Res. 521
Turkey	None	28 Jun 2025: stablecoin withdrawals capped at USD 3,000 a day and USD 50,000 a month	Decree 32, MASAK No. 29
United States	None	1 Jan 2026: 1% tax on remittances funded with cash	IRC §4475
Germany	None	1 Jan 2025: reporting threshold raised from EUR 12,500 to EUR 50,000	BGBl. 2024 I 411

Panel B: no limit, no change found

Mexico (DOF 1991), Colombia (R.E. 1/2018), Czech Republic (TFEU 63), Poland (Dz.U. 2024/1131), Romania (BNR Reg. 4/2005), United Kingdom (Finance Act 1987 s. 68), Australia (RBA 1989), Canada (CBSA), Switzerland (Federal Council 2016), Japan (FEFTA Art. 1)

Country	Rule	In force	Source
Panel A: a rule on individuals' stablecoin purchases or transfers			
Turkey	Stablecoin withdrawals, transfers to other platforms included: USD 3,000 a day and USD 50,000 a month; 48-hour wait	28 Jun 2025	MASAK No. 29
Brazil	Stablecoin trades and self-custody wallet transfers are foreign-exchange operations; crypto payments abroad via non-bank providers capped at USD 100,000	2 Feb 2026	BCB Res. 521
South Korea	Transfers of KRW 1 million or more between exchanges carry sender and recipient data	Throughout	Decree Art. 10-10
Philippines	Retail access to unregistered offshore exchanges not allowed	30 Jan 2026	BSP M-2026-003
Argentina	Banks and payment providers may not offer or facilitate crypto operations for clients	2022, 2023	BCRA A 7506, A 7759
DE, CZ, PL, RO	Only authorised issuers may offer e-money tokens; platforms may not trade non-compliant stablecoins (Tether is not on ESMA's register); self-hosted wallet transfers above EUR 1,000 need an ownership check	30 Jun, 30 Dec 2024	MiCA Art. 48, Reg. 2023/1113
Canada	Platforms may offer only stablecoins whose issuer gave regulators an undertaking (none found for Tether)	31 Dec 2024	CSA Notice 21-333
Japan	A foreign stablecoin is handled only if its issuer is licensed under an equivalent foreign law; transfers capped at JPY 1 million each	Throughout	Cabinet Office Ord. Art. 30
Mexico	Banks and fintech firms may not offer crypto exchange, transfer or custody to clients	Throughout	Circular 4/2019
Panel B: duties on platforms only, no limit on individuals			
United States (GENIUS Act), United Kingdom (MLR Part 7A), Australia (ASIC INFO 225), Switzerland (FINMA 06/2024), Taiwan (FSC registration rules), South Africa, Colombia			